

Appendix C. Humanoid robot price, BOM and total ownership cost data

This appendix reports the robot-price, bill-of-materials (BOM), cost-category and total-ownership-cost inputs used in the humanoid robot parcel-sorting economic model. All monetary values are retained in the currency used in the input workbook unless explicitly stated. Missing or unconfirmed values are reported as N/A. Tables are split into portrait-format blocks to preserve readability and avoid horizontal overflow.

Table C1: Workbook overview for humanoid robot cost appendix

Summary values taken from the dashboard sheet of the robot cost workbook.

Item	Value
Workbook purpose	Track 2026 public prices and cost/TCO buckets for humanoid robots relevant to parcel sorting ROI.
Data status	Mix of official public prices, media-reported prices, and enterprise estimates. Quote vendors before final investment decision.
Models covered	18
Lowest public price RMB	9,998
Median public price RMB	180,000
Highest public price RMB	1,800,000

Table C2: Dashboard summary of robot models and five-year TCO

This table summarizes the robot models retained for cost comparison. TCO values include deployment, operating and depreciation assumptions from the workbook.

ID	Company	Model	Price mid RMB	5-year TCO RMB	Comment
1	Noetix Robotics	Bumi / Xiaobumi	9,998	468,316.94	Low-cost/dev class
2	Unitree	R1 Air	35,280	506,998.4	Low-cost/dev class
3	Unitree	R1	42,480	518,014.4	Low-cost/dev class
4	Unitree	R1 domestic entry	33,400	504,122	Low-cost/dev class
5	Booster Robotics	Booster K1	29,900	498,767	Low-cost/dev class
6	AstriBot	T1	89,900	590,567	Low-cost/dev class
7	Unitree	G1	97,200	601,736	Low-cost/dev class
8	Unitree	H2	215,280	782,398.4	Potential pilot class
9	Unitree	H1	648,000	1,444,460	High-cost enterprise class
10	Unitree	H2 Plus	720,000	1,554,620	High-cost enterprise class
11	AgiBot / Zhiyuan	Lingxi X2 Youth	98,000	602,960	Low-cost/dev class
12	AgiBot / Zhiyuan	Yuanzheng A2 Youth	183,000	733,010	Potential pilot class
13	AgiBot / Zhiyuan	Genie G1	450,000	1,141,520	High-cost enterprise class
14	UBTECH	Walker S2	676,800	1,488,524	High-cost enterprise class
15	Fourier Intelligence	GR-1	1,584,000	2,876,540	Very high-cost enterprise/RaaS
16	Agility Robotics	Digit	1,800,000	3,207,020	Very high-cost enterprise/RaaS
17	Tesla	Optimus	180,000	728,420	Potential pilot class

Table C3: Robot price database: model identity and price status

Public-price status separates official prices, media-reported prices, enterprise estimates and benchmark-only values.

ID	Company	Model	Segment	Region	Price mid RMB	Price status	Quality
1	Noetix Robotics	Bumi / Xiaobumi	Consumer / education	China	9,998	Public sale price	High
2	Unitree	R1 Air	Entry humanoid	China	35,280	Official shop / product page	High
3	Unitree	R1	Entry humanoid	China	42,480	Official shop / product page	High
4	Unitree	R1 domestic entry	Entry humanoid	China	33,400	Media-reported domestic range	Medium
5	Booster Robotics	Booster K1	Education / research	China	29,900	Public launch price	High
6	AstriBot	T1	Industrial / general	China	89,900	Media-reported starting price	Medium
7	Unitree	G1	Research / development	China	97,200	Official shop price	High
8	Unitree	H2	Full-size humanoid	China	215,280	Official shop price	High
9	Unitree	H1	Full-size humanoid	China	648,000	Official shop indicative price	High
10	Unitree	H2 Plus	Full-size humanoid	China	720,000	Official shop price	High
11	AgiBot / Zhiyuan	Lingxi X2 Youth	Education / interaction	China	98,000	Public sale price	High
12	AgiBot / Zhiyuan	Yuanzheng A2 Youth	Interaction / service	China	183,000	Public sale / promo range	High
13	AgiBot / Zhiyuan	Genie G1	High-end humanoid	China	450,000	Public sale price	High

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ID	Company	Model	Segment	Region	Price mid RMB	Price status	Quality
14	UBTECH	Walker S2	Industrial humanoid	China	676,800	Enterprise estimate / quote	Low
15	Fourier Intelligence	GR-1	Industrial / re-search	China	1,584,000	Partner quote	Medium
16	Agility Robotics	Digit	Warehouse logistics	United States	1,800,000	Industry estimate / RaaS	Medium
17	Tesla	Optimus	Benchmark / not for sale	United States	180,000	Target price; not purchasable	Low
18	Tesla	Optimus Gen 2 BOM benchmark	BOM benchmark	United States / China supply chain	637,200	BOM estimate range	Medium

Table C4: Robot price database: original price ranges and capability notes

Original prices are retained to show whether the public value was reported in RMB, USD or another currency.

ID	Model	Original low	Original high	Curr.	RMB low	RMB high	Use case / capability
1	Bumi / Xiaobumi	9,998	9,998	RMB	9,998	9,998	94cm, ~12kg, 21+ DOF; education/companion/developer use
2	R1 Air	4,900	4,900	USD	35,280	35,280	Compact humanoid, 20 DOF, tax/shipping excluded
3	R1	5,900	5,900	USD	42,480	42,480	Compact humanoid, 26 DOF, tax/shipping excluded
4	R1 domestic entry	26,900	39,900	RMB	26,900	39,900	Domestic 2026 mass-market price reference
5	Booster K1	29,900	29,900	RMB	29,900	29,900	1m, ~19kg, developer/education platform
6	T1	89,900	89,900	RMB	89,900	89,900	Industrial humanoid entering sub-RMB 100k range
7	G1	13,500	13,500	USD	97,200	97,200	Compact 35kg humanoid, 23-43 joint motors, optional dexterous hand
8	H2	29,900	29,900	USD	215,280	215,280	Full-size humanoid listed on Unitree shop
9	H1	90,000	90,000	USD	648,000	648,000	Full-size H1, contact sales for real price
10	H2 Plus	100,000	100,000	USD	720,000	720,000	Higher-end H2 Plus listed on Unitree shop
11	Lingxi X2 Youth	98,000	98,000	RMB	98,000	98,000	1.3m-class interaction/development humanoid
12	Yuanzheng A2 Youth	168,000	198,000	RMB	168,000	198,000	1.7m-class interaction/service humanoid
13	Genie G1	450,000	450,000	RMB	450,000	450,000	High-end embodied intelligence product
14	Walker S2	68,000	120,000	USD	489,600	864,000	Industrial factory/logistics model with hot-swap battery
15	GR-1	220,000	220,000	USD	1,584,000	1,584,000	Full-size research/rehab/industrial pilot humanoid
16	Digit	250,000	250,000	USD	1,800,000	1,800,000	Warehouse bin/tote handling; Amazon pilots; often RaaS
17	Optimus	20,000	30,000	USD	144,000	216,000	Future scaled-production price target only
18	Optimus Gen 2 BOM benchmark	46,000	131,000	USD	331,200	943,200	China supply-chain cost vs ex-China supply-chain cost estimate

Table C5: Robot price database: source URLs and source notes

URLs are retained as plain text links for auditability. Some enterprise prices are estimates or quotes and should be verified before procurement decisions.

ID	Company	Model	Source URL	Notes
1	Noetix Robotics	Bumi / Xiaobumi	https://noetixrobotics.com/news/media/213	Not a logistics-grade sorter; price useful as lower-bound consumer benchmark.
2	Unitree	R1 Air	https://www.unitree.com/mobile/R1/	Official page lists R1 Air at \$4,900.
3	Unitree	R1	https://www.unitree.com/mobile/R1/	Official page lists R1 at \$5,900; domestic media also reports lower local entry prices.
4	Unitree	R1 domestic entry	https://www.chinadailyasia.com/article/634870	China Daily Asia reports starting price RMB 26,900; earlier Reuters/media reported ~RMB 39,900.
5	Booster Robotics	Booster K1	https://www.chinadailyasia.com/article/634870	China Daily Asia reports K1 series starting price RMB 29,900.
6	Astriobot	T1	https://www.chinadailyasia.com/article/634870	Reported starting price; full configuration may be higher.

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ID	Company	Model	Source URL	Notes
7	Unitree	G1	https://shop.unitree.com/collections/humanoid-robot	Official shop lists G1 at \$13,500; Chinese media reports G1 base near RMB 85k.
8	Unitree	H2	https://shop.unitree.com/collections/humanoid-robot	Official shop lists H2 at \$29,900.
9	Unitree	H1	https://shop.unitree.com/collections/humanoid-robot	Official shop shows \$90,000 and says contact for real price.
10	Unitree	H2 Plus	https://shop.unitree.com/collections/humanoid-robot	Official shop lists H2 Plus at \$100,000.
11	AgiBot / Zhiyuan	Lingxi X2 Youth	https://news.cnyes.com/news/id/6115374	Reported as AgiBot’s cheapest humanoid model at RMB 98,000.
12	AgiBot / Zhiyuan	Yuanzheng A2 Youth	https://news.cnyes.com/news/id/6115374	Cnyes reports RMB 198,000 list and RMB 30,000 limited discount; Sina reports RMB 168,000.
13	AgiBot / Zhiyuan	Genie G1	https://finance.sina.com.cn/tech/roll/2025-08-18/doc-infmmiy1961486.shtml	Sina/Jiemian reports Genie G1 at RMB 450,000.
14	UBTECH	Walker S2	https://blog.robozaps.com/b/ubtech-walker-s-review	UBTECH enterprise quotes vary; public exact price not disclosed.
15	Fourier Intelligence	GR-1	https://www.roboticscenter.ai/store/product/fourier-gr1/	SVRC partner quote lists \$220,000; official Fourier page requires quote.
16	Agility Robotics	Digit	https://blog.robozaps.com/b/agility-robotics-digit-review	Estimated outright price; Agility often sells service/subscription deployment.
17	Tesla	Optimus	https://standardbots.com/blog/tesla-robot	No confirmed retail price; use only as forecast/benchmark, not current purchase quote.
18	Tesla	Optimus Gen 2 BOM benchmark	https://wallstreetcn.com/articles/3774890	WallstreetCN cites Tesla BOM \$46k using China supply chain vs \$131k using US/non-China supply chain.

Table C6: BOM benchmark components and cost values

BOM values combine public/component benchmarks from BofA, Roland Berger and WallstreetCN-cited component estimates.

Benchmark	Source	Component	Share	Cost low	Cost high	Curr.	RMB low	RMB high
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Linear actuator integration	0.27	9,450	9,450	USD	68,040	68,040
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Rotary actuator integration	0.24	8,400	8,400	USD	60,480	60,480
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Dexterous hand / end effector	0.19	6,650	6,650	USD	47,880	47,880
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	AI chip / AI computing	0.1	3,500	3,500	USD	25,200	25,200
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Vision system	0.04	1,400	1,400	USD	10,080	10,080
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Battery	0.02	700	700	USD	5,040	5,040
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Heat management system	0.02	700	700	USD	5,040	5,040
BofA typical humanoid BOM (China-made components, end-2025)	BofA Global Research	Other body / structural / connectors	0.12	4,200	4,200	USD	30,240	30,240
Roland Berger 2026 entry-level component model	Roland Berger	Motion – actuator	N/A	1,265	1,265	USD	9,108	9,108
Roland Berger 2026 entry-level component model	Roland Berger	Hand/end-effector system	N/A	333	333	USD	2,397.6	2,397.6
Roland Berger 2026 entry-level component model	Roland Berger	Motion – other	N/A	342	342	USD	2,462.4	2,462.4
Roland Berger 2026 entry-level component model	Roland Berger	Energy & charging	N/A	380	380	USD	2,736	2,736
Roland Berger 2026 entry-level component model	Roland Berger	Skeleton & structural components	N/A	820	820	USD	5,904	5,904
Roland Berger 2026 entry-level component model	Roland Berger	Compute & connectivity	N/A	345	345	USD	2,484	2,484
Roland Berger 2026 entry-level component model	Roland Berger	Perception system	N/A	166	166	USD	1,195.2	1,195.2

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Benchmark	Source	Component	Share	Cost low	Cost high	Curr.	RMB low	RMB high
Roland Berger 2026 entry-level component model	Roland Berger	Others	N/A	230	230	USD	1,656	1,656
Roland Berger 2026 advanced component model	Roland Berger	Motion – actuator	N/A	4,000	4,000	USD	28,800	28,800
Roland Berger 2026 advanced component model	Roland Berger	Hand/end-effector system	N/A	1,400	1,400	USD	10,080	10,080
Roland Berger 2026 advanced component model	Roland Berger	Motion – other	N/A	1,072	1,072	USD	7,718.4	7,718.4
Roland Berger 2026 advanced component model	Roland Berger	Energy & charging	N/A	845	845	USD	6,084	6,084
Roland Berger 2026 advanced component model	Roland Berger	Skeleton & structural components	N/A	1,950	1,950	USD	14,040	14,040
Roland Berger 2026 advanced component model	Roland Berger	Compute & connectivity	N/A	800	800	USD	5,760	5,760
Roland Berger 2026 advanced component model	Roland Berger	Perception system	N/A	465	465	USD	3,348	3,348
Roland Berger 2026 advanced component model	Roland Berger	Others	N/A	440	440	USD	3,168	3,168
Unitree G1 teardown / media estimate	WallstreetCN / Zhineng Auto	Total material cost / BOM	N/A	41,600	41,600	RMB	41,600	41,600
Unitree G1 teardown / media estimate	WallstreetCN / Zhineng Auto	Core joints	N/A	27,500	27,500	RMB	27,500	27,500
Tesla Optimus supply-chain estimate	WallstreetCN / Morgan Stanley estimate cited	BOM with Chinese supply chain	N/A	46,000	46,000	USD	331,200	331,200
Tesla Optimus supply-chain estimate	WallstreetCN / Morgan Stanley estimate cited	BOM with US/non-China supply chain	N/A	131,000	131,000	USD	943,200	943,200

Table C7: BOM benchmark notes and source links

Notes describe the component-level assumptions behind each cost estimate.

Benchmark	Component	Notes	Source URL
BofA typical humanoid BOM (China-made components, end-2025)	Linear actuator integration	Assumes 14 linear actuators based on planetary roller screws.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Rotary actuator integration	Assumes 16 rotary actuators with harmonic reducers.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Dexterous hand / end effector	Assumes hand with 6 DOF.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	AI chip / AI computing	Onboard compute / AI chip.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Vision system	One depth camera and one LiDAR assumption.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Battery	Battery pack.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Heat management system	Thermal management.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
BofA typical humanoid BOM (China-made components, end-2025)	Other body / structural / connectors	Residual percentage to make total 100%.	https://institute.bankofamerica.com/content/dam/transformation/humanoid-robots.pdf
Roland Berger 2026 entry-level component model	Motion – actuator	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Hand/end-effector system	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Motion – other	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf

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Benchmark	Component	Notes	Source URL
Roland Berger 2026 entry-level component model	Energy & charging	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Skeleton & structural components	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Compute & connectivity	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Perception system	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 entry-level component model	Others	Entry-level single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Motion – actuator	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Hand/end-effector system	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Motion – other	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Energy & charging	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Skeleton & structural components	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Compute & connectivity	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Perception system	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Roland Berger 2026 advanced component model	Others	Advanced single robot component cost.	https://content.rolandberger.com/hubfs/07_presse/Roland_Berger_Humanoid_Robots_2026.pdf
Unitree G1 teardown / media estimate	Total material cost / BOM	Media article states G1 material cost RMB 41,600.	https://wallstreetcn.com/articles/3774890
Unitree G1 teardown / media estimate	Core joints	Media article states G1 core joints RMB 27,500.	https://wallstreetcn.com/articles/3774890
Tesla Optimus supply-chain estimate	BOM with Chinese supply chain	Media article cites \$46k with Chinese supply-chain components.	https://wallstreetcn.com/articles/3774890
Tesla Optimus supply-chain estimate	BOM with US/non-China supply chain	Media article cites \$131k if excluding Chinese components.	https://wallstreetcn.com/articles/3774890

Table C8: Robot TCO model: upfront deployment components

Upfront deployment cost adds the bare price, estimated parcel-handling adaptation, barcode/scanning, safety and integration assumptions.

ID	Company	Model	Price mid RMB	Hardware BOM RMB	End effector	Barcode	Safety	Integration	Upfront cost
1	Noetix Robotics	Bumi / Xi-aobumi	9,998	5,998.8	999.8	5,000	299.94	1,999.6	18,297.34
2	Unitree	R1 Air	35,280	21,168	3,528	5,000	1,058.4	7,056	51,922.4
3	Unitree	R1	42,480	25,488	4,248	5,000	1,274.4	8,496	61,498.4
4	Unitree	R1 domestic entry	33,400	20,040	3,340	5,000	1,002	6,680	49,422
5	Booster Robotics	Booster K1	29,900	17,940	2,990	5,000	897	5,980	44,767
6	Astribot	T1	89,900	53,940	8,990	5,000	2,697	17,980	124,567
7	Unitree	G1	97,200	58,320	9,720	5,000	2,916	19,440	134,276
8	Unitree	H2	215,280	129,168	21,528	5,000	6,458.4	43,056	291,322.4
9	Unitree	H1	648,000	388,800	64,800	5,000	19,440	129,600	866,840
10	Unitree	H2 Plus	720,000	432,000	72,000	5,000	21,600	144,000	962,600
11	AgiBot / Zhiyuan	Lingxi X2 Youth	98,000	58,800	9,800	5,000	2,940	19,600	135,340
12	AgiBot / Zhiyuan	Yuanzheng A2 Youth	183,000	109,800	18,300	5,000	5,490	36,600	248,390
13	AgiBot / Zhiyuan	Genie G1	450,000	270,000	45,000	5,000	13,500	90,000	603,500
14	UBTECH	Walker S2	676,800	406,080	67,680	5,000	20,304	135,360	905,144
15	Fourier Intelligence	GR-1	1,584,000	950,400	158,400	5,000	47,520	316,800	2,111,720
16	Agility Robotics	Digit	1,800,000	1,080,000	180,000	5,000	54,000	360,000	2,399,000
17	Tesla	Optimus	180,000	108,000	18,000	5,000	5,400	36,000	244,400

Table C9: Robot TCO model: annual operating components*Annual operating costs are modelled per robot and are used to convert robot price into annual economic cost.*

ID	Model	Maintenance	Software	Ops personnel	Electricity	Downtime loss
1	Bumi / Xiaobumi	799.84	28,800	22,500	2,304	36,000
2	R1 Air	2,822.4	28,800	22,500	2,304	36,000
3	R1	3,398.4	28,800	22,500	2,304	36,000
4	R1 domestic entry	2,672	28,800	22,500	2,304	36,000
5	Booster K1	2,392	28,800	22,500	2,304	36,000
6	T1	7,192	28,800	22,500	2,304	36,000
7	G1	7,776	28,800	22,500	2,304	36,000
8	H2	17,222.4	28,800	22,500	2,304	36,000
9	H1	51,840	28,800	22,500	2,304	36,000
10	H2 Plus	57,600	28,800	22,500	2,304	36,000
11	Lingxi X2 Youth	7,840	28,800	22,500	2,304	36,000
12	Yuanzheng A2 Youth	14,640	28,800	22,500	2,304	36,000
13	Genie G1	36,000	28,800	22,500	2,304	36,000
14	Walker S2	54,144	28,800	22,500	2,304	36,000
15	GR-1	126,720	28,800	22,500	2,304	36,000
16	Digit	144,000	28,800	22,500	2,304	36,000
17	Optimus	14,400	28,800	22,500	2,304	36,000

Table C10: Robot TCO model: depreciation, residual value and five-year TCO*Five-year TCO is the cumulative ownership-cost benchmark used for comparing low-cost, pilot-class and enterprise-class humanoids.*

ID	Model	Residual value	Annual depreciation	Annual operating cost	5-year TCO	TCO / purchase chase	Status / comment
1	Bumi / Xiaobumi	1,999.6	3,259.55	90,403.84	468,316.94	46.84	Low-cost/dev class
2	R1 Air	7,056	8,973.28	92,426.4	506,998.4	14.37	Low-cost/dev class
3	R1	8,496	10,600.48	93,002.4	518,014.4	12.19	Low-cost/dev class
4	R1 domestic entry	6,680	8,548.4	92,276	504,122	15.09	Low-cost/dev class
5	Booster K1	5,980	7,757.4	91,996	498,767	16.68	Low-cost/dev class
6	T1	17,980	21,317.4	96,796	590,567	6.57	Low-cost/dev class
7	G1	19,440	22,967.2	97,380	601,736	6.19	Low-cost/dev class
8	H2	43,056	49,653.28	106,826.4	782,398.4	3.63	Potential pilot class
9	H1	129,600	147,448	141,444	1,444,460	2.23	High-cost enterprise class
10	H2 Plus	144,000	163,720	147,204	1,554,620	2.16	High-cost enterprise class
11	Lingxi X2 Youth	19,600	23,148	97,444	602,960	6.15	Low-cost/dev class
12	Yuanzheng A2 Youth	36,600	42,358	104,244	733,010	4.01	Potential pilot class
13	Genie G1	90,000	102,700	125,604	1,141,520	2.54	High-cost enterprise class
14	Walker S2	135,360	153,956.8	143,748	1,488,524	2.20	High-cost enterprise class
15	GR-1	316,800	358,984	216,324	2,876,540	1.82	Very high-cost enterprise/RaaS
16	Digit	360,000	407,800	233,604	3,207,020	1.78	Very high-cost enterprise/RaaS
17	Optimus	36,000	41,680	104,004	728,420	4.05	Potential pilot class

Table C11: Robot cost-category map: definitions and default estimation*The cost-category map links the model's requested cost buckets to concrete data items and estimation rules.*

Cost bucket	What it includes	Default estimation	Use in ROI	Caveat
Bare machine / robot body	Public robot purchase price or manufacturer quote; excludes tax/shipping/custom integration	Use mid public price from Robot_Price_DB	Main capex; compare against annual labour saving	Public prices often refer to developer models, not industrial configurations.
End effector	Dexterous hands, gripper, suction cup, parcel handling tooling, tactile sensors	End_effector_upgrade_rate x hardware price	Essential for parcels; default hands may not be enough for soft packs and irregular parcels	Often optional and not included in base robot price.
Vision system	Depth camera, LiDAR, RGB cameras, industrial camera, lighting, OCR vision	Included in base if manufacturer includes; otherwise allocate from BOM benchmarks	Required for parcel localisation and bin-picking	Can be much higher if adding industrial cameras and lighting.
Barcode/scanning system	Fixed scanner, handheld/robot-mounted 1D/2D scanner, OCR/DWS integration	Scanner_system_base_RMB	Critical for express sorting; humanoid vision alone does not replace barcode workflows	Usually not part of humanoid price.
Mobility system	Leg actuators, joint modules, reducers, motors, screws, encoders, bearings, servo drives	Part of base robot; compare to BOM actuator shares	Determines whether robot can move between stations or only stand at risk. fixed workcell	Largest cost bucket, also main maintenance

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Cost bucket	What it includes	Default estimation	Use in ROI	Caveat
Safety system	E-stop, collision avoidance, protective covers, fences, warning lights, risk assessment, insurance enablement	Safety_system_rate_base x hardware price	Required if operating near workers; affects deployment permission	Often hidden in integration budget.
Software cost	Fleet software, robot SDK, navigation updates, cloud, monitoring, VLA/LLM subscriptions	Software_license_base_RMB per robot/year	Can determine whether robot is Opex-heavy even if hardware is cheap	Vendors may bundle first-year software.
Integration cost	WMS/WCS/TMS integration, workflow redesign, simulation, training, acceptance testing	Integration_rate_base x hardware price	Often decides whether pilot can scale	Highly site-specific; real quote needed.
Maintenance cost	Preventive maintenance, spare parts, battery replacement, joint/gear wear, service contract	Maintenance_rate_base x hardware price/year	Heavy-duty night sorting may push maintenance upward	Warehouse robotics benchmark; humanoid data still limited.
Operations personnel	Robot technician, trainer, remote operator, fleet supervisor	Ops_engineer_annual_cost_RMB / Robots_per_ops_engineer	If one person manages few robots, ROI worsens sharply	Supervision need is often underestimated.
Electricity	Charging energy and battery swap station power	Power_kW x hours/day x days/year x tariff	Small compared with labour, but useful for complete TCO	Power varies by task intensity and payload.
Downtime loss	Lost throughput, backup labour, missed cut-off, service delay and recovery costs	Downtime_hours_per_year x Downtime_cost_per_hour_RMB	Important for express logistics because cut-off times matter	Hard to estimate without site data.
Depreciation	Annual accounting cost of robot, integration and accessories net of residual value	Straight-line depreciation over Useful_life_years	Turns capex into annual cost to compare with annual labour cost	Rapid price declines can shorten economic life.
Residual value	Expected resale/scrap value at end of useful life	Residual_rate x hardware price	Higher residual improves payback; low residual raises risk	Very uncertain in 2026 due to rapid price declines.

Table C12: Robot cost-category map: data collection and low/base/high logic

These fields specify what should be collected from vendors or site pilots to replace modelling proxies.

Cost bucket	Recommended data to collect	Low/base/high logic
Bare machine / robot body	Model, configuration, hand option, battery option, warranty, tax, shipping	If no quote, use low/high public price range
End effector	Hand model, payload, grip success rate, parcel types, replacement frequency	5% / 10% / 20% of hardware price
Vision system	Sensor bill, camera resolution, lighting, perception software	BofA vision 4%; Roland perception entry \$166/advanced \$465
Barcode/scanning system	Scanner unit price, mounting, SDK/WMS integration, scan success rate	RMB 500 / RMB 5,000 / RMB 20,000
Mobility system	DOF, actuator count, motor/reducer supplier, payload, walking speed, uptime	BofA linear+rotary actuator = 51%; Wall-streetCN says joint modules ~half
Safety system	CE/GB compliance, safety certification, risk assessment cost, near-miss events	1% / 3% / 8% of hardware price
Software cost	Annual license, API usage, cloud compute, model updates, cybersecurity	RMB 7,200 / RMB 28,800 / RMB 72,000 per year
Integration cost	Engineering days, WMS API, safety testing, line downtime during installation	10% / 20% / 50% of hardware price
Maintenance cost	Service contract, warranty, replacement schedule, critical spares, MTBF	5% / 8% / 12% of hardware price/year
Operations personnel	Technician salary, robots per technician, teleoperation need	Depends on supervision ratio
Electricity	kWh per hour, operating hours, electricity tariff	Editable power and hours
Downtime loss	Downtime hours, parcel throughput/hour, penalty/backup labour cost/hour	Editable low/base/high
Depreciation	Useful life, residual value, capex basis	3 / 5 / 7 years
Residual value	Second-hand market, supplier buyback, spare-part value	5% / 20% / 35%

Table C13: Robot cost assumptions and low/base/high values

Core cost assumptions are explicitly separated from the robot price database so they can be modified in sensitivity analysis.

Parameter	Base	Low	High	Unit
USD_to_RMB	7.2	7	7.4	RMB per USD
EUR_to_RMB	7.8	7.5	8.1	RMB per EUR
Maintenance_rate_base	0.08	0.05	0.12	% of hardware price/year
Software_license_base_RMB	28,800	7,200	72,000	RMB/year
Integration_rate_base	0.2	0.1	0.5	% of hardware price
Safety_system_rate_base	0.03	0.01	0.08	% of hardware price
Scanner_system_base_RMB	5,000	500	20,000	RMB upfront
End_effector_upgrade_rate	0.1	0.05	0.2	% of hardware price
Ops_engineer_annual_cost_RMB	180,000	120,000	300,000	RMB/year
Robots_per_ops_engineer	8	4	15	robots/person
Electricity_tariff_RMB_per_kWh	0.8	0.6	1.2	RMB/kWh
Power_kW_base	0.6	0.3	1.2	kW

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Parameter	Base	Low	High	Unit
Operating_hours_per_day	16	8	22	hours/day
Operating_days_per_year	300	250	350	days/year
Downtime_hours_per_year	120	40	300	hours/year
Downtime_cost_per_hour_RMB	300	100	1,000	RMB/hour
Useful_life_years	5	3	7	years
Residual_rate	0.2	0.05	0.35	% of initial hardware price
BOM_proxy_rate_of_sale_price	0.6	0.4	0.8	%

Table C14: Robot cost assumptions: notes and source links

Source notes include URLs when the assumption was calibrated from public warehouse-robotics or humanoid-cost information.

Parameter	Notes / source URL
USD_to_RMB	Editable conversion assumption; keep explicit rather than hidden FX.
EUR_to_RMB	Editable conversion assumption.
Maintenance_rate_base	Warehouse robotics guide suggests 5-12% of hardware cost annually: https://robotomated.com/learn/cost/warehouse-robot-cost-guide
Software_license_base_RMB	Proxy converted from \$1,000-\$10,000 per robot/year: https://robotomated.com/learn/cost/warehouse-robot-cost-guide
Integration_rate_base	Site-specific WMS/WCS integration, tooling, safety testing; editable proxy.
Safety_system_rate_base	Safety barriers, E-stop, warning lights, insurance enablement; not usually included in robot price.
Scanner_system_base_RMB	Barcode/OCR scanner add-on for logistics; industrial modules range from low-cost modules to rugged scanners.
End_effector_upgrade_rate	For parcel sorting, gripper/suction/dexterous-hand upgrade or replacement; editable proxy.
Ops_engineer_annual_cost_RMB	One robotics technician/engineer fully loaded annual cost assumption.
Robots_per_ops_engineer	Supervision ratio; increase after deployment maturity.
Electricity_tariff_RMB_per_kWh	Editable industrial/commercial tariff assumption.
Power_kW_base	Proxy based on 0.5-2 kWh per charge patterns and humanoid battery/runtime data; adjust by model/task.
Operating_hours_per_day	For logistics night/day shift scenario.
Operating_days_per_year	Editable site utilisation assumption.
Downtime_hours_per_year	Maintenance/unplanned downtime proxy.
Downtime_cost_per_hour_RMB	Throughput loss + backup labour + delay penalty proxy.
Useful_life_years	Straight-line depreciation assumption.
Residual_rate	Residual value highly uncertain because humanoid prices are falling quickly.
BOM_proxy_rate_of_sale_price	If no public BOM: estimate hardware BOM as share of public selling price; disclose as proxy only.